

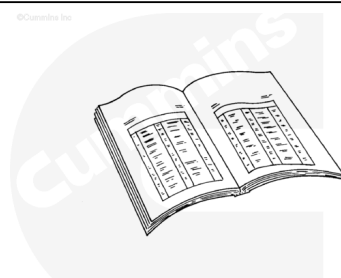
Trainings ▾

Preparatory Steps

Automotive with CM875, CM870 and CM570

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



ck800wa

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Note : If the exhaust gas recirculation (EGR) cooler is being replaced due to an internal coolant leak, the EGR differential pressure tubes **must** be replaced. Refer to Procedure 011-026 in Section 11.

(/qs3/pubsys2/xml/en/procedures/35/35-011-026-tr.html)

Note : Brush away all loose dirt from around the area of the air-handling connections to reduce the possibility of contamination of the interior of the engine.

Note : Use protective caps from the ISM Clean Care Kit, Part Number 4919404, and the OEM Clean Care Kit, Part Number 4919425, to cover open connection points.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Remove the exhaust gas pressure sensor tube. Refer to Procedure 011-027 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-027-tr.html)
- Remove the exhaust gas recirculation cooler coolant lines. Refer to Procedure 011-031 in Section 11.

(/qs3/pubsys2/xml/en/procedures/35/35-011-031-tr.html)

- Remove the EGR differential pressure sensor tubes. Refer to Procedure 011-026 in Section 11.

(/qs3/pubsys2/xml/en/procedures/35/35-011-026-tr.html)

- Remove the EGR connection tube. Refer to Procedure 011-025 in Section 11.

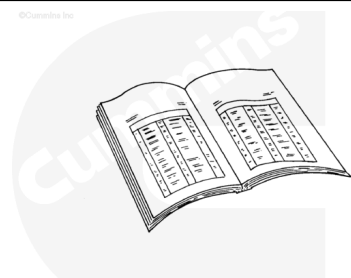
(/qs3/pubsys2/xml/en/procedures/35/35-011-025-tr.html)

- Remove the EGR mixer. Refer to Procedure 011-021 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-021-tr.html)

Automotive with CM876

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



ck800wa

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Note : If the EGR cooler is being replaced due to a malfunction that caused an internal coolant leak, the crankcase breather element must be replaced. Refer to Procedure 003-019 in Section 3
(/qs3/pubsys2/xml/en/procedures/101/101-003-019-tr.html)

Note : If the EGR cooler is being replaced due to a malfunction that caused an internal coolant leak, the EGR differential pressure sensor tubes are to be inspected for coolant. If coolant is found, replace the EGR differential pressure sensor tubes. Refer to Procedure 011-026 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-026-tr.html)

Note : Brush away any loose dirt from around the area of the air handling connections to reduce the

possibility of contamination of the interior of the engine.

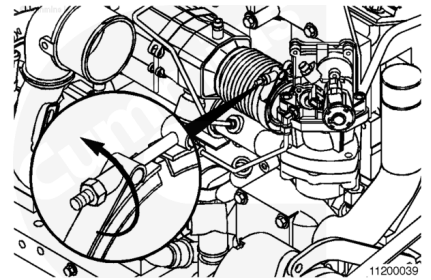
Note : Use protective caps from the ISM Clean Care Kit, Part Number 4919404, and the OEM Clean Care Kit, Part Number 4919425, to cover open connection points.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>)
- Remove the air intake pipe to the turbocharger. Refer to the OEM service manual.
- Remove the exhaust gas recirculation cooler coolant lines. Refer to Procedure 011-031 in Section 11. (</qs3/pubsys2/xml/en/procedures/35/35-011-031-tr.html>)
- Remove the charge-air cooler pipe between the turbocharger and charge-air cooler. Refer to the OEM service manual.
- Remove the exhaust gas recirculation cooler coolant lines. Refer to Procedure 011-031 in Section 11. (</qs3/pubsys2/xml/en/procedures/35/35-011-031-tr.html>)
- Remove the EGR differential pressure sensor tubes. Refer to Procedure 011-026 in Section 11. (</qs3/pubsys2/xml/en/procedures/35/35-011-026-tr.html>)
- Remove the EGR connection tube and EGR venturi tube. Refer to Procedure 011-025 in Section 11. (</qs3/pubsys2/xml/en/procedures/35/35-011-025-tr.html>)
- Remove the EGR mixer. Refer to Procedure 011-021 in Section 11. (</qs3/pubsys2/xml/en/procedures/35/35-011-021-tr.html>)
- Remove the aftertreatment fuel injector coolant return line. Refer to Procedure 011-051 in Section 11. (</qs3/pubsys2/xml/en/procedures/101/101-011-051-tr.html>)
- Remove the turbocharger coolant return line. Refer to Procedure 010-041 in Section 10. (</qs3/pubsys2/xml/en/procedures/35/35-010-041-tr.html>)

Remove

Automotive with CM875, CM870 and CM570

Loosen the V-Band clamp between the EGR cooler and the EGR valve. When removing the V-Band clamp, do **not** damage the EGR connection tube.



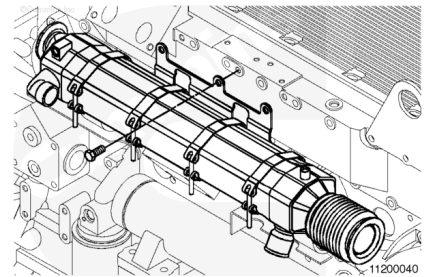
⚠ CAUTION ⚠

The straps on the EGR cooler are not serviceable. Adjusting or removing the straps can damage the EGR cooler.

Remove the five capscrews that attach the EGR cooler to the intake manifold.

Remove the EGR cooler assembly from the engine.

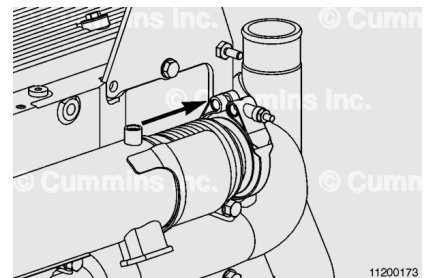
Use protective caps from the ISM Clean Care Kit, Part Number 4919404, to cover open connection points on both the EGR cooler and EGR plumbing.



Automotive with CM876

Loosen the V-Band clamp between the EGR cooler and the exhaust manifold.

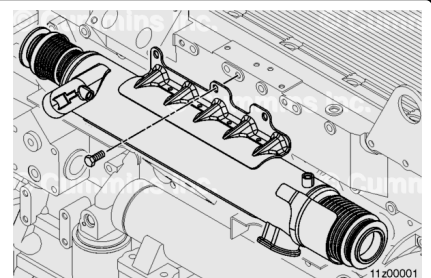
To prevent damage to the EGR cooler bellows, place the V-Band clamp over the exhaust manifold flange.



Remove the five capscrews that attach the EGR cooler to the intake manifold.

Remove the EGR cooler assembly from the engine.

Use protective caps from the ISM Clean Care Kit, Part Number 4919404, to cover open connection points on both the EGR cooler and EGR plumbing.

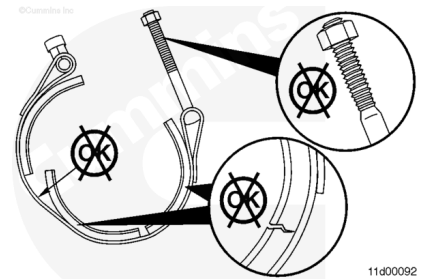


Clean and Inspect for Reuse

Automotive with CM875, CM870 and CM570

Inspect the V-Band clamp threads for damage. Replace the clamp if damage is found.

Inspect the hose clamps for damage. Replace if damage is found.

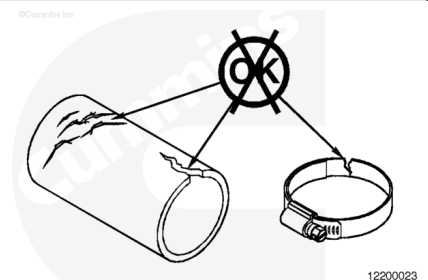


Inspect the hose clamp(s) for cracks or other damage.

Replace hose clamp(s) if cracks or other damage is found.

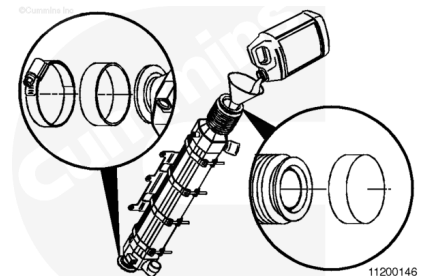
Inspect the hose(s) for cracks or other damage.

Replace the hose(s) if cracks or other damage is found.



⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



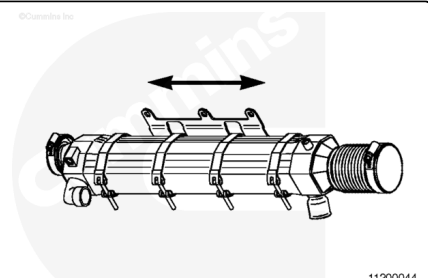
Cap the outlet end of the EGR cooler with parts supplied in the EGR Cooler Leak Test Kit, Part Number 3164843.

Completely fill the EGR cooler with safety solvent.

Cap the gas inlet end of the EGR cooler with parts supplied in EGR Cooler Leak Test Kit, Part Number 3164843.

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Lay the EGR cooler on its side.

Allow the EGR cooler to soak for approximately 20 minutes.

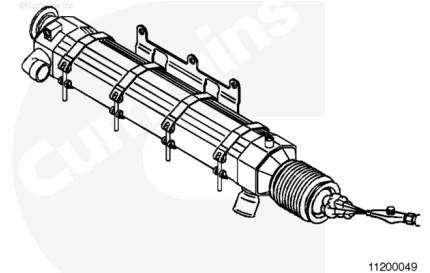
Drain approximately 25 percent of the liquid from the cooler and recap.

Properly dispose of the liquid.

Shake the cooler by hand, from end to end, for approximately 30 seconds.

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Do not use chlorinated water to clean or rinse the EGR cooler. Chlorides are very corrosive to the stainless alloys internal and external to the EGR cooler.

- Drain the safety solvent from the EGR cooler and properly dispose of the liquid.
- Use compressed air to dry the inside of the EGR cooler and remove any loose debris or particles.
- The EGR cooler can be cleaned again if necessary.
- Cap the outlet end of the EGR cooler with parts supplied in the EGR Cooler Leak Test Kit, Part Number 3164843.
- Completely fill the EGR cooler with water.
- Cap the gas inlet end of the EGR cooler with parts supplied in the EGR Cooler Leak Test Kit, Part Number 3164843.
- Shake the cooler by hand, from end to end, for approximately 30 seconds.
- Drain the water from the EGR cooler and properly dispose of the liquid.
- Use compressed air to dry the inside of the EGR cooler.

Clean the EGR valve with safety solvent where the EGR cooler connection mates to the EGR valve. Refer to Procedure 011-022 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-022-tr.html)

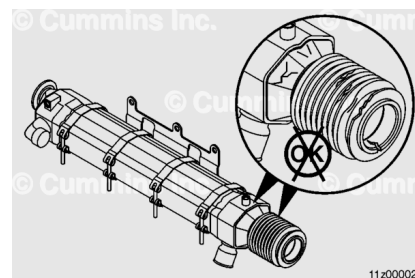
Clean the EGR cooler connection with safety solvent and dry with compressed air. Refer to Procedure 011-025 in Section 11 (/qs3/pubsys2/xml/en/procedures/35/35-011-025-tr.html).

Inspect the EGR valve where the EGR cooler connection mates to the EGR valve. Replace the EGR valve if cracks or other damage is found. Refer to Procedure 011-022 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-022-tr.html)

Inspect the EGR cooler connection. Replace the EGR cooler connection if cracks or other damage is found. Refer to Procedure 011-025 in Section 11.

(/qs3/pubsys2/xml/en/procedures/35/35-011-025-tr.html)

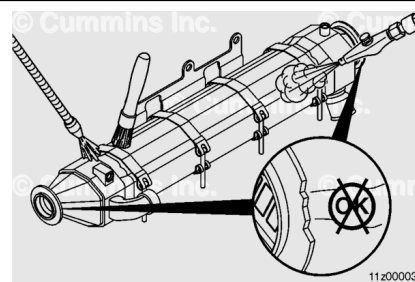
Replace the EGR cooler if cracks or other damage is found.



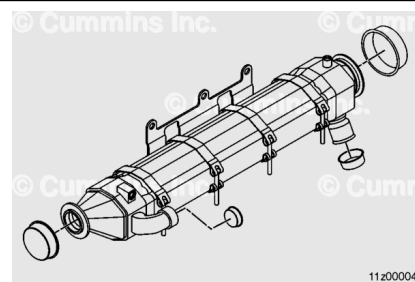
⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

Clean the mounting surfaces with safety solvent to remove any coolant deposits or debris.

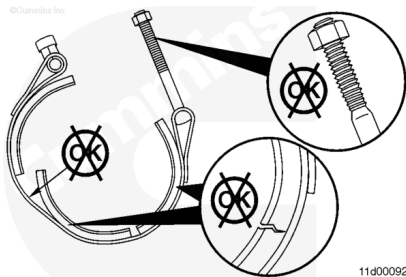


After the parts are clean and dry, cover open connection points with protective caps from the ISM Clean Care Kit, Part Number 4919404, or with heavy tape, if needed.

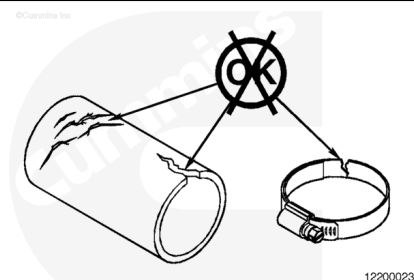


Automotive with CM876

Inspect the V-Band clamp threads for damage. Replace the V-Band clamp if damage is found.

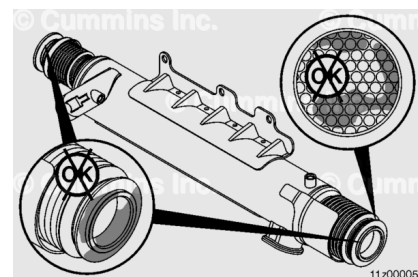


Inspect the hose clamp(s) for cracks or other damage.
 Replace hose clamp(s) if cracks or other damage is found.
 Inspect the hose(s) for cracks or other damage.
 Replace the hose(s) if cracks or other damage is found.



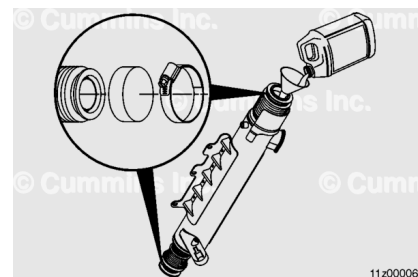
Inspect the exhaust gas inlet, exhaust gas outlet, and exhaust gas tubes for coolant deposits and note their location for reference during the pressure test procedure.

Note : Reference the following service bulletin for more information regarding inspection of the EGR cooler.
 Reference the EGR Cooler Reuse Guidelines for X Series Engines, Bulletin 2883594.



⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Cap the outlet end of the EGR cooler with the plastic caps supplied in the EGR Cooler Leak Test Kit, Part Number 4919424.

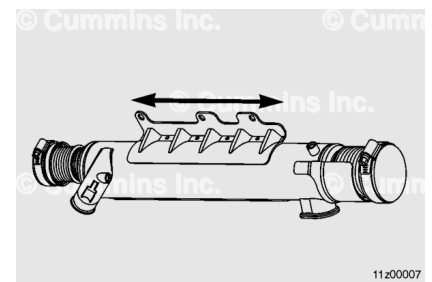
Completely fill the EGR cooler with safety solvent.

Cap the gas inlet end of the EGR cooler with the plastic cap supplied in the EGR Cooler Leak Test Kit, Part Number 4919424.

⚠ WARNING ⚠

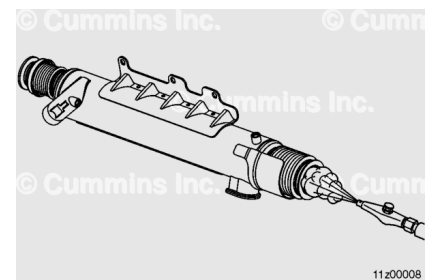
When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

- Lay the EGR cooler on its side.
- Allow the EGR cooler to soak for approximately 20 minutes.
- Drain approximately 25 percent of the liquid from the cooler and recap.
- Properly dispose of the liquid.
- Shake the cooler by hand, from end to end, for approximately 30 seconds.



⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

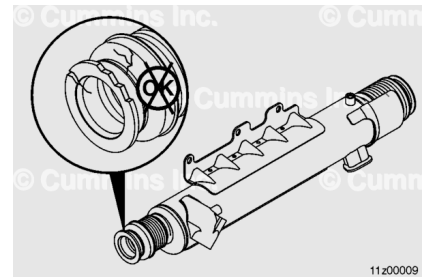
Do not use chlorinated water to clean or rinse the EGR cooler. Chlorides are very corrosive to the stainless alloys internal and external to the EGR cooler.

- Drain the safety solvent from the EGR cooler and properly dispose of the liquid.
- Use compressed air to dry the inside of the EGR cooler and remove any loose debris or particles.
- The EGR cooler can be cleaned again, if necessary.
- Cap the outlet end of the EGR cooler with parts supplied in the EGR Cooler Leak Test Kit, Part Number 4919424.
- Completely fill the EGR cooler with water.
- Cap the gas inlet end of the EGR cooler with the plastic cap supplied in the EGR Cooler Leak Test Kit, Part Number 4919424.

- Shake the cooler by hand, from end to end, for approximately 30 seconds.
- Drain the water from the EGR cooler and properly dispose of the liquid.
- Use compressed air to dry the inside of the EGR cooler.
- Clean the exhaust manifold with safety solvent, where the EGR cooler connection attaches to the exhaust manifold. Refer to Procedure 011-007 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-007-tr.html)
- Inspect the exhaust manifold where the EGR cooler connection attaches to the exhaust manifold. Replace the exhaust manifold if cracks or other damage is found. Refer to Procedure 011-007 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-007-tr.html)

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris or dirt can cause personal injury.



Clean the EGR cooler exhaust gas outlet with safety solvent.

Dry with compressed air.

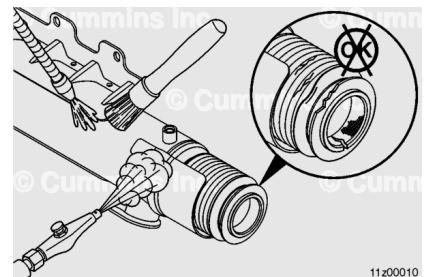
Inspect the EGR cooler exhaust gas outlet for signs of fretting, cracks, or other damage.

Replace the EGR cooler if damage is found. Refer to Procedure 011-025 in Section 11.

(/qs3/pubsys2/xml/en/procedures/35/35-011-025-tr.html)

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris or dirt can cause personal injury.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

Clean the EGR cooler bellows with safety solvent and dry with compressed air.

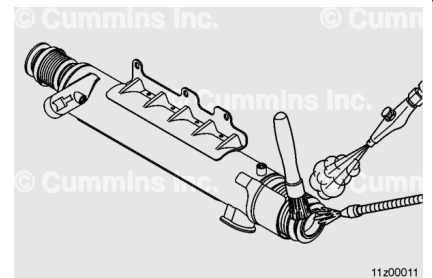
Inspect the EGR cooler bellows for cracks, dents, bending, or other damage.

Replace the EGR cooler if cracks, dents, bending, or other damage is found.

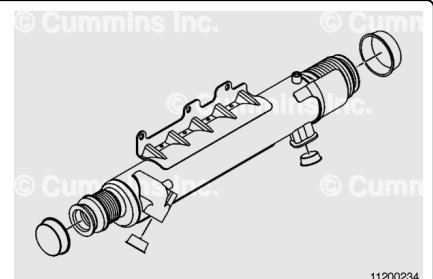
⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

Clean the mounting surfaces with safety solvent to remove any coolant deposits or debris.



After the parts are clean and dry, cover open connection points with protective caps from the ISM Clean Care Kit, Part Number 4919404, or with heavy tape, if needed.

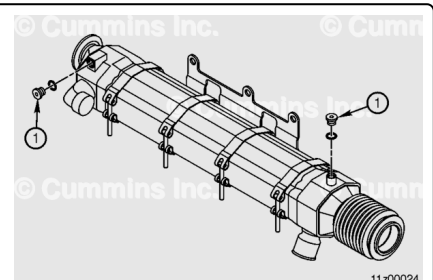


Pressure Test

Automotive with CM875, CM870 and CM570

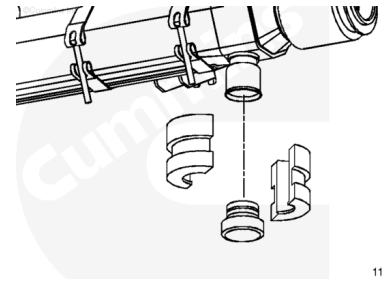
Clean the EGR cooler using the steps found in the Clean and Inspect for Reuse section of this procedure, if **not** already completed.

Install threaded plugs, part number 3040814, from the EGR Cooler Leak Test Kit, Part Number 3164843, into the vent ports (1) of the EGR cooler.



Cap the coolant supply port of the EGR cooler with plug, Part Number 3164844, and collar, Part Number 3164736, from the EGR Cooler Leak Test Kit, Part Number 3164843.

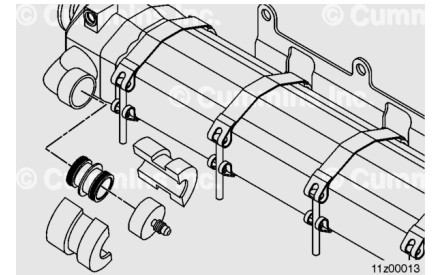
Secure with a hose clamp, Part Number 3056787.



Note : Jumper tube, Part Number 3683068, and o-ring, Part Number 3049221, from the EGR cooler are used during this step.

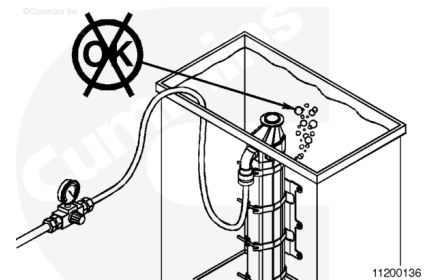
Cap the coolant return port of the EGR cooler with jumper tube, Part Number 3683068, cap, part number 3164737, and collar, Part Number 3164736, from the EGR Cooler Leak Test Kit, Part Number 3164843.

Secure with hose clamp, Part Number 3056787.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Note : The EGR cooler must be properly cleaned of soot accumulation and other debris before submerging it in the test tank.

Note : Air Pressure Regulator Kit, Part Number 3164231, or equivalent, is suitable for use during this step.

Install the air pressure regulator, Part Number 3164231, or equivalent, onto the plug, Part Number 3164737, in the EGR cooler coolant return port.

Make sure the air pressure regulator is closed, and connect compressed air to the pressure regulator.

Completely submerge the EGR cooler in room temperature water. Make sure to submerge the cooler vertically, with the EGR cooler connection down, so that air can **not** be left in the open (gas) side of the cooler.

Apply 206 kPa [30 psi] of air pressure to the cooler.

Inspect for bubbles escaping from uncapped gas ports of the EGR cooler.

Note : Bubbles coming from a damaged EGR cooler will be very small or will not be a steady stream.

Verify that bubbles are **not** a result of loose fittings or trapped air.

If bubbles are **not** observed, the EGR cooler is usable.

Make sure the pressure regulator is closed.

Disconnect the air pressure from the EGR cooler assembly.

Carefully vent the trapped air from the EGR cooler assembly.

Remove the parts supplied in the EGR Cooler Leak Test Kit, Part Number 3164843.

If the EGR cooler is **not** usable, inspect the turbocharger turbine housing and rear EGR connection tube for any signs of coolant deposits. Coolant deposits can be identified as dried white deposits coating the inside of the turbine housing or EGR connection tube.

If coolant deposits are found in the turbocharger turbine housing, inspect the turbocharger for progressive damage.

Refer to Procedure 010-033 in Section 10.

(/qs3/pubsys2/xml/en/procedures/35/35-010-033-tr.html)

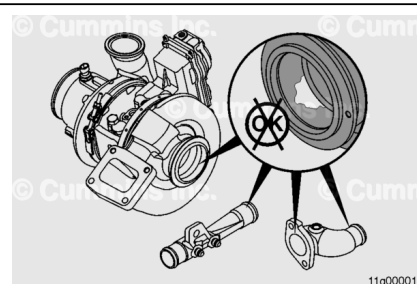
If coolant deposits are found in the EGR connection tube, inspect the EGR connection tube(s) and EGR mixer for reuse.

Refer to Procedure 011-025 in Section 11.

(/qs3/pubsys2/xml/en/procedures/35/35-011-025-tr.html)

If the EGR cooler is being replaced due to damage that caused an internal coolant leak:

- Replace the EGR differential pressure sensor tubes. Refer to Procedure 011-026 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-026-tr.html)
- Check for EGR differential pressure sensor fault codes. If fault codes are found, follow the appropriate troubleshooting tree, and replace the EGR differential pressure sensor, if required. Use the following procedure in the ISM CM875 Electronic Control System Troubleshooting and Repair Manual, Bulletin 4021477, or ISM CM870 with CM570 Electronic Control System Troubleshooting and Repair Manual, Bulletin 4021381. Refer to Procedure 019-370 in Section 19.
(/qs3/pubsys2/xml/en/procedures/88/88-019-370.html)



Note : If damage resulted in a hydraulic lock and the coolant level reached the intake manifold, it may be necessary to clean the charge air cooler. Refer to the equipment manufacturer service information.

Note : If damage resulted in a hydraulic lock and the coolant level reached the intake manifold, inspect the air compressor. Refer to Procedure 012-014 in Section 12. (</qs3/pubsys2/xml/en/procedures/35/35-012-014-tr.html>)

Note : If damage resulted in a hydraulic lock, it may be necessary to remove the injectors and evacuate coolant from the cylinder(s). Refer to Procedure 006-026 in Section 6.

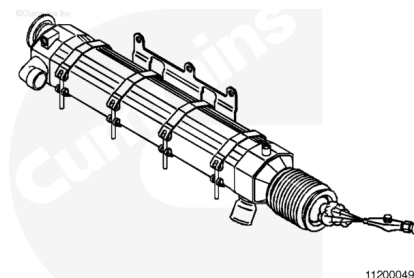
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⚠ WARNING ⚠

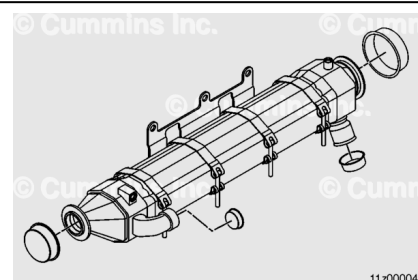
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Drain all of the water from the gas side of the EGR cooler before reuse.

Dry the cooler with compressed air.



After testing is completed and parts are dry, cover open connection points with protective caps from the ISM Clean Care Kit, Part Number 4919404, or with heavy tape, if needed.

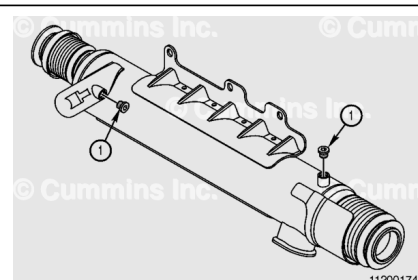


Automotive with CM876

Clean the EGR cooler using the steps found in the Clean and Inspect for Reuse section of this procedure, if **not** already completed.

Note : This test is performed with the EGR cooler removed from the engine.

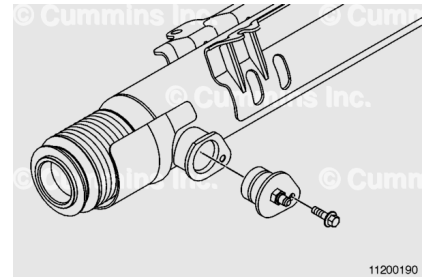
Install the threaded plugs (1), Part Number 3040814, from the EGR Cooler Leak Test Kit, Part Number 4919424, into the coolant line port and vent port of the EGR cooler.



Install the plug, Part Number 4918559, with the o-ring, Part Number 3049221, from the EGR Cooler Leak Test Kit, Part Number 4919424, into the coolant inlet port of the EGR cooler.

Secure the plug with a M8 x 1.25 capscrew.

Torque Value: 21 n•m [186 in-lb]

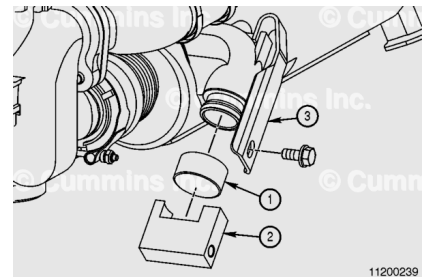


Note : Jumper tube, Part Number 3683068, o-ring, Part Number 3049221, and bracket, Part Number 4966430, from the EGR cooler are used during this step

Install the cap (1), Part Number 4918450, and clamp (2), Part Number 4919359, from the EGR Cooler Leak Test Kit, Part Number 4919424 over jumper tube, Part Number 3683068.

Secure with bracket (3), Part Number 4966430, onto the EGR cooler coolant outlet port with a M8 x 1.25 capscrew.

Torque Value: 21 n•m [186 in-lb]



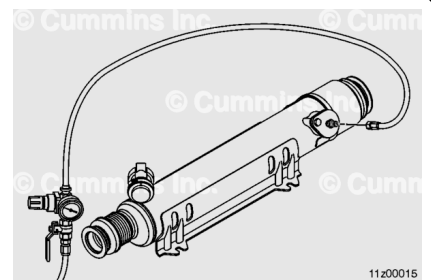
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Note : Air pressure regulator kit, Part Number 3164231, or equivalent, is suitable for use during this step.

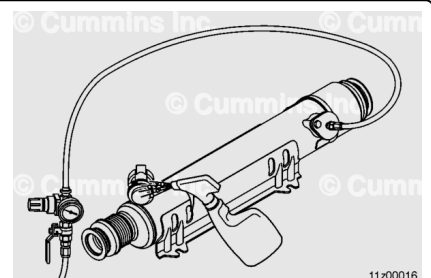
Install the air pressure regulator, Part Number 3164231, or equivalent, on the plug, Part Number 4918559, in the EGR cooler coolant supply port.

Install a ball-type shutoff valve onto the supply side of the air pressure regulator, to allow the air supply to be shut off during testing.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Note : The test must be performed when the EGR cooler has reached ambient temperature. Do not apply air pressure to a hot EGR cooler.

Make sure the air pressure regulator and ball-type shutoff valve are closed and connect a compressed air supply to the pressure regulator.

Apply 25 PSI (172 kPa) of air pressure to the EGR cooler.

Inspect for air escaping from the EGR cooler or pressure test assembly.

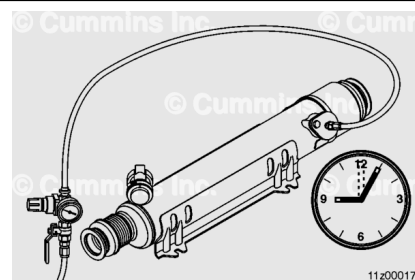
Check for leaks by spraying the exterior of the EGR cooler and all lines and fittings with a mixture of mild soap and water.

Repair leaking fittings or lines.

Replace the EGR cooler if external leaks, caused by cracks or other damage, are found on the EGR cooler.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Shut the air supply off to the air pressure regulator, Part Number 3164231, using the ball valve.

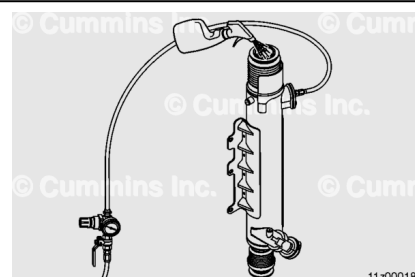
Disconnect the compressed air supply from the air pressure regulator.

Record the time the compressed air supply was shut off and removed from the air pressure regulator.

Record the pressure shown on the air pressure regulator gauge.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

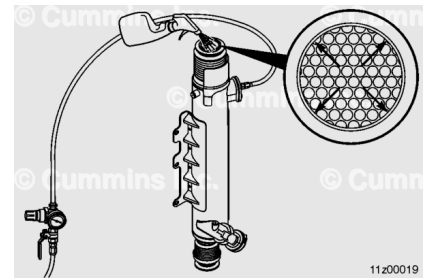


Place the EGR cooler in a vertical position with the exhaust gas inlet bellows pointing upward.

Spray the exhaust inlet bulkhead and gas tubes with a mixture of mild soap and water.

Gently flex the exhaust gas inlet bellows so that the inside corners of the bulkhead can be sprayed as well.

Note : It is important to wet the entire bulkhead with the mixture of mild soap and water.



Allow the EGR cooler to remain pressurized for 15 minutes.

Gently flex the bellows and inspect the bulkhead and gas tubes for the continuous formation of bubbles.

Measure and record the air pressure.

If the measured pressure loss in the EGR cooler is more than 103 kPa [15 psi] in 15 minutes, or if there is a continuous formation of bubbles near the gas tubes and bulkhead, the EGR cooler is **not** usable.

Note : Reference the following service bulletin for more information regarding inspection of the EGR cooler. Reference the EGR Cooler Reuse Guidelines for X Series Engines, Bulletin 2883594.

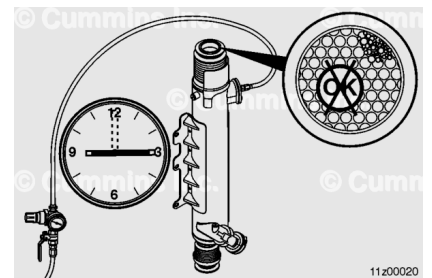
Note : The pressure test is meant to identify large leaks, and the inspection with mild soap and water is meant to identify small leaks. Do not use small losses observed during the pressure test only to evaluate the EGR cooler for reuse, as they are often associated with issues with the leak test equipment.

Note : Stationary bubbles that form on the soot deposits or in the gas tubes are not an indication of a leak.

Note : The continuous formation of bubbles is an indication that the EGR cooler is **not** usable.

Note : If the EGR cooler is **not** usable based on a loss of pressure, carefully inspect the air pressure regulator, air lines, ball valves, o-rings, and other components for leaks. A mixture of mild soap and water may be sprayed on the assembly to aid in leak detection.

Note : If the EGR cooler is **not** usable based on a loss of pressure, it can be necessary to evaluate the EGR cooler pressure testing assembly for leaks before replacing the EGR cooler. Assorted fittings or a known good EGR cooler can be used for this test.



Make sure the pressure regulator is closed.

Disconnect the air pressure from the EGR cooler assembly.

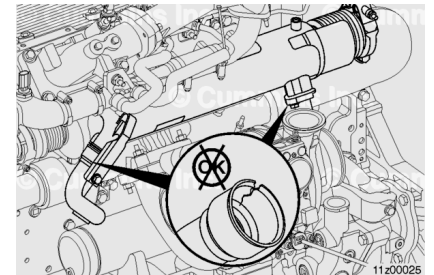
Carefully vent the trapped air from the EGR cooler assembly.

Remove the components installed from the EGR Cooler Leak Test Kit, Part Number 4919424.



If the EGR cooler is usable, inspect the EGR cooler coolant lines, couplings, o-rings, and grommets for cracks or other damage.

Replace the necessary components if cracks or other damage is found. Refer to Procedure 011-031 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-031-tr.html)



If the EGR cooler is **not** reusable, inspect the turbocharger turbine housing and rear EGR connection tube for any signs of coolant deposits.

Coolant deposits can be identified as dried white deposits coating the inside of the turbine housing.

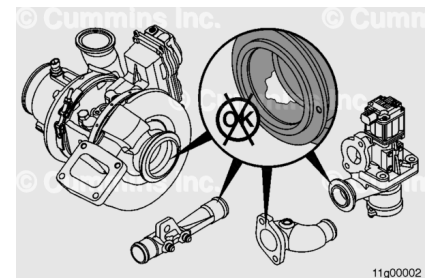
If coolant deposits are found in the turbocharger turbine housing, check for turbocharger related fault codes. If turbocharger related fault codes are found, inspect the turbocharger. Refer to Procedure 010-033 in Section 10.
(/qs3/pubsys2/xml/en/procedures/35/35-010-033-tr.html)

If coolant deposits are found in the rear EGR connection tube, inspect the EGR valve for progressive damage. Refer to Procedure 011-022 in Section 11
(/qs3/pubsys2/xml/en/procedures/35/35-011-022-tr.html).

If coolant deposits are found in the EGR valve, inspect the EGR venturi and front EGR connection tubes for progressive damage. Refer to Procedure 011-025 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-025-tr.html)

If the EGR cooler is being replaced due to damage that caused an internal coolant leak:

- Clean the aftertreatment fuel injector, regardless if coolant deposits are present or **not**. Refer to Procedure 011-042 in Section 11. (/qs3/pubsys2/xml/en/procedures/101/101-011-042-tr.html)
- Replace the crankcase breather element. Refer to Procedure 003-019 in Section 3.



(/qs3/pubsys2/xml/en/procedures/101/101-003-019-tr.html)

- Inspect the crankcase pressure sensor. Use the following procedure in the ISX CM871 and ISM CM876 Electronic Control System Troubleshooting Manual, Bulletin 4021560. Refer to Procedure 019-445 in Section 19. (/qs3/pubsys2/xml/en/procedures/00/00-019-445.html)
- Inspect the EGR differential pressure sensor tubes for coolant. If coolant is found, replace the EGR differential pressure sensor tubes. Refer to Procedure 011-026 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-026-tr.html)
- Check for EGR differential pressure sensor fault codes. If fault codes are found, follow the appropriate troubleshooting tree, and replace the EGR differential pressure sensor, if required. Use the following procedure in the ISX CM871 and ISM CM876 Electronic Control System Troubleshooting and Repair Manual, Bulletin 4021560. Refer to Procedure 019-370 in Section 19. (/qs3/pubsys2/xml/en/procedures/16/16-019-370.html)
- If coolant entered the exhaust system, the aftertreatment system can be recovered. Refer to Procedure 014-013 in Section 14. (/qs3/pubsys2/xml/en/procedures/101/101-014-013-tr.html)

Note : If damage resulted in a hydraulic lock and the coolant level reached the intake manifold, it may be necessary to clean the charge air cooler. Refer to the equipment manufacturer service information.

Note : If damage resulted in a hydraulic lock and the coolant level reached the intake manifold, inspect the air compressor. Refer to Procedure 012-014 in Section 12. (/qs3/pubsys2/xml/en/procedures/35/35-012-014-tr.html)

Note : If damage resulted in a hydraulic lock, it may be necessary to remove the injectors and evacuate coolant from the cylinder(s). Refer to Procedure 006-026 in Section 6.

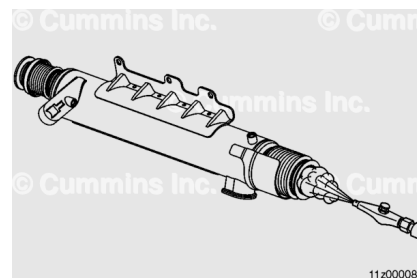
(/qs3/pubsys2/xml/en/procedures/35/35-006-026-tr.html)

⚠ WARNING ⚠

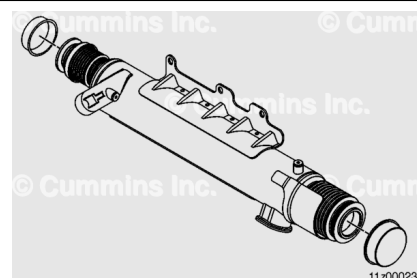
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Drain all the water from the gas side of the EGR cooler before reuse.

Dry the cooler with compressed air.



After testing is completed and parts are dry, cover open connection points with protective caps from the ISM Clean Care Kit, Part Number 4919404, or with heavy tape, if needed.



Install

Automotive with CM875, CM870 and CM570

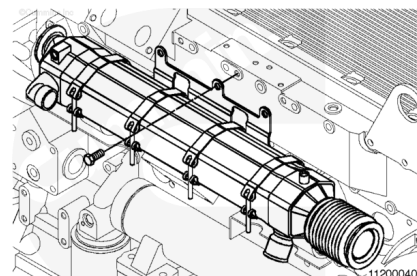
Remove any protective caps or heavy tape.

Loosely install the bottom two mounting capscrews and install the EGR cooler onto them.

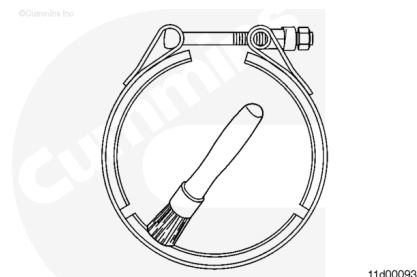
Install the top three capscrews and tighten all five capscrews.

Torque Value: 20 n•m [177 in-lb]

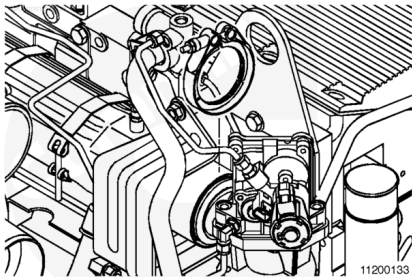
Install a new EGR cooler connection gasket with the adhesive side on the piloted end of the EGR cooler connection, where it connects with the EGR valve.



Before installing the V-Band clamp, coat the clamping surface with high temperature anti-seize compound.

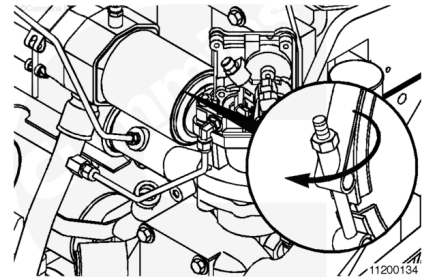


Install the V-Band clamps between the EGR valve and EGR cooler connection.



Tighten the V-Band clamp between the EGR cooler connection and the EGR valve.

Torque Value: 17 n•m [150 in-lb]



Automotive with CM876

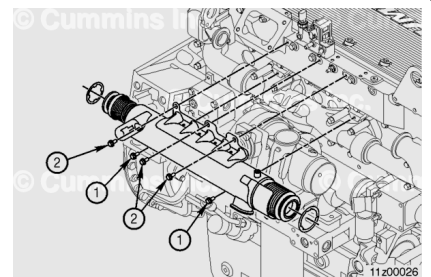
Remove any protective caps or heavy tape.

Loosely install the bottom two mounting capscrews (1) and install the EGR cooler onto them.

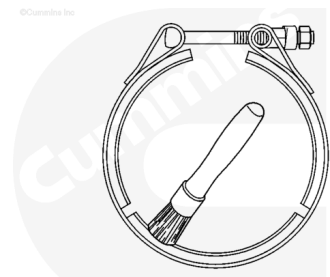
Install the top three capscrews (2) and tighten all five capscrews.

Torque Value: 20 n•m [177 in-lb]

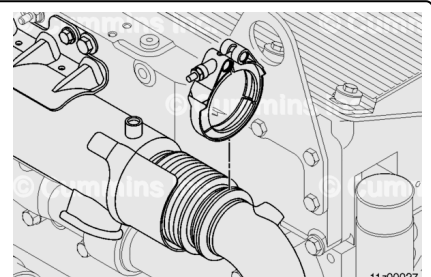
Install new EGR cooler connection gaskets with the adhesive side on the piloted end of each joint.



Before installing the V-Band clamp, coat the clamping surface with high temperature anti-seize compound.

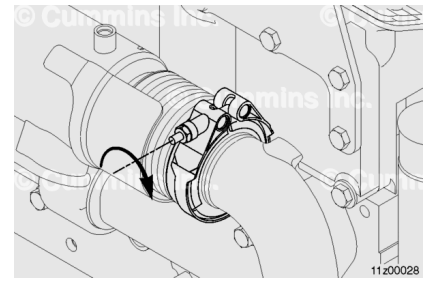


Install the V-Band clamp between the EGR valve and the exhaust manifold.



Tighten the V-Band clamps.

Torque Value: 17 n•m [150 in-lb]



Finishing Steps

Automotive with CM875, CM870 and CM570

Note : If the EGR cooler is being replaced due to an internal coolant leak, the EGR differential pressure tubes **must** be replaced. Refer to Procedure 011-026 in Section 11. (</qs3/pubsys2/xml/en/procedures/35/35-011-026-tr.html>)



- Install the EGR mixer. Refer to Procedure 011-021 in Section 11. (</qs3/pubsys2/xml/en/procedures/35/35-011-021-tr.html>)
- Install the EGR connection tube. Refer to Procedure 011-025 in Section 11. (</qs3/pubsys2/xml/en/procedures/35/35-011-025-tr.html>)
- Install the EGR differential pressure sensor tubes. Refer to Procedure 011-026 in Section 11. (</qs3/pubsys2/xml/en/procedures/35/35-011-026-tr.html>)
- Install the exhaust gas pressure sensor tube. Refer to Procedure 011-027 in Section 11. (</qs3/pubsys2/xml/en/procedures/35/35-011-027-tr.html>)
- Install the EGR cooler connection heat shield. Refer to Procedure 011-032 in Section 11. (</qs3/pubsys2/xml/en/procedures/35/35-011-032-tr.html>)
- Install the EGR cooler coolant lines. Refer to Procedure 011-031 in Section 11. (</qs3/pubsys2/xml/en/procedures/35/35-011-031-tr.html>)
- Fill engine cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008->

018-tr.html)

- Start the engine, check for leaks, and verify proper operation.

Automotive with CM876

Note : If the EGR cooler is being replaced due to a malfunction that caused an internal coolant leak, the crankcase breather element **must** be replaced. Refer to Procedure 003-019 in Section 3
(/qs3/pubsys2/xml/en/procedures/101/101-003-019-tr.html)



Note : If the EGR cooler is being replaced due to a malfunction that caused an internal coolant leak, the EGR differential pressure sensor tubes are to be inspected for coolant. If coolant is found, replace the EGR differential pressure sensor tubes. Refer to Procedure 011-026 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-026-tr.html)

Note : If a malfunction resulted in oil, **excessive fuel**, or **excessive black smoke** entering the exhaust system, the aftertreatment system **must** be inspected. Consult the Aftertreatment Diesel Oxidation Catalyst and Aftertreatment Diesel Particulate Filter Reuse Guidelines, Bulletin 4021600.
(/qs3/pubsys2/xml/en/bulletin/4021600.html)

Note : If a malfunction resulted in coolant entering the exhaust system, the aftertreatment system can be recovered. Refer to Procedure 014-013 in Section 14.
(/qs3/pubsys2/xml/en/procedures/101/101-014-013.html)

- Install the turbocharger aftertreatment injector lines. Refer to Procedure 011-051 in Section 11.
(/qs3/pubsys2/xml/en/procedures/101/101-011-051-tr.html)
- Install the turbocharger coolant return line. Refer to Procedure 010-041 in Section 10.
(/qs3/pubsys2/xml/en/procedures/35/35-010-041-tr.html)
- Install the EGR mixer. Refer to Procedure 011-021 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-021-tr.html)
- Install the EGR connection tube and EGR venturi tube. Refer to Procedure 011-025 in Section 11.

(/qs3/pubsys2/xml/en/procedures/35/35-011-025-tr.html)

- Install the EGR differential pressure sensor tubes. Refer to Procedure 011-026 in Section 11.

(/qs3/pubsys2/xml/en/procedures/35/35-011-026-tr.html)

- Install the EGR cooler coolant lines. Refer to Procedure 011-031 in Section 11.

(/qs3/pubsys2/xml/en/procedures/35/35-011-031-tr.html)

- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Start the engine, check for leaks, and verify proper operation.